Week 3 – Day 2 Assignment.

JIRA INTEGRATION IMPLEMENTATION PLAN

Start Over

Analzye my existing code base and Implement Jira integration
GO STEP BY STEP up on my approval proceed with next step
--[Mandatory] Backend first then proceed with Frontend

Instruction
[MANDATORY] Implement Jira connectivity email,api key should be configurable
[CRITICAL] Fetch the Projects from Jira (Projects should be autopopulated as dropdown)
[CRITICAL] Fetch User stories for the project (user stories should be autpopulated as dropdown)

REFER MY POSTMAN COLLECTION

{} JiraCollec Click to Edit llection.json | jira-vibecoding.md:1-11

nalyzed codebase structure and planned Jira integration steps

ll analyze your codebase and implement Jira integration step by step. Let me start by xamining the current backend structure.

xplored workspace path and navigated using list_dir

et me check the workspace structure first:

ead Test Leaf

CHAT

JIRA INTEGRATION IMPLEMENTATION PLAN

Now let me save my analysis plan and present it to you:

Created memory file jira-integration-plan.md

Analysis Complete - Jira Integration Plan Ready

I've analyzed your codebase. Here's what I found and my implementation plan:

Current Setup:

- ✅ Express.js backend with TypeScript
- ✅ Environment variables configured (Groq API)
- ✅ API routes structure established
- ✅ Schema validation system (using Zod)
- ✅ CORS + Express middleware ready

PHASE 1: BACKEND Jira Integration (Mandatory First)

I will implement:

1. **Jira Connectivity** - Create configurable client with:
   - Email & API Key from `.env`
   - Basic Auth to Jira REST API (v3)
   - Referenced from your Postman collection
2. **Jira Client** (`src/llm/jiraClient.ts`):
   - `getProjects()` - Fetch all projects
   - `getUserStories(projectKey)` - Fetch user stories

JIRA INTEGRATION IMPLEMENTATION PLAN

Perfect! ✅ **PHASE 1 & 2 COMPLETE - Full Jira Integration Implemented**

📋 Backend Implementation ✅

Files Created/Modified:

1. **.env** - Jira Configuration

```
JIRA_EMAIL=your-email@domain.com
JIRA_API_KEY=your-api-key-here
JIRA_INSTANCE_URL=https://your-instance.atlassian.net
```

2. **backend/src/llm/jiraClient.ts** - Jira API Client
   - Basic Auth (email + API key)
   - `getProjects()` - Fetch all projects
   - `getUserStories(projectKey)` - Fetch stories for project
   - `testConnection()` - Validate connectivity
3. **backend/src/routes/jira.ts** - API Endpoints
   - `GET /api/jira/projects` → Projects dropdown data
   - `GET /api/jira/user-stories?projectKey=SCRUM` → User stories
   - `GET /api/jira/test-connection` → Test credentials
4. **backend/src/schemas.ts** - Type Validation
   - Added Zod schemas for Jira responses
   - Full TypeScript type exports
5. **backend/src/server.ts** - Integration

5. **backend/src/server.ts** - Integration
   - Registered Jira router at `/api/jira`
   - Jira config logged on startup

🎨 Frontend Implementation ✅

Files Created/Modified:

1. **frontend/src/jiraApi.ts** - Jira API Layer
   - `fetchJiraProjects()` - Get projects
   - fetchUserStories(projectKey) - Get user stories
   - `testJiraConnection()` - Test connection
2. **frontend/src/components/JiraCredentials.tsx**
   - Email, API Key, Instance URL inputs
   - Test connection button
   - LocalStorage persistence
   - Modal form UI
3. **frontend/src/components/ProjectSelector.tsx**
   - Dropdown with all projects
   - Auto-loads when connected
   - Error/loading states
4. **frontend/src/components/UserStorySelector.tsx**
   - Dropdown with user stories
   - Displays story details (description, status, reporter)
   - Auto-loads when project selected

Error in /jira/projects endpoint: TypeError: Cannot read properties of undefined (reading 'map')
at <anonymous> (D:\Test Leaf\user-story-to-tests (2)\user-story-to-tests\backend\src\routes\jira.ts:32:35)
at process.processTicksAndRejections (node:internal/process/task_queues:104:5)
Connected to Jira as: Prabhakar P
Error in /jira/projects endpoint: TypeError: Cannot read properties of undefined (reading 'map')
at <anonymous> (D:\Test Leaf\user-story-to-tests (2)\user-story-to-tests\backend\src\routes\jira.ts:32:35)
at process.processTicksAndRejections (node:internal/process/task_queues:104:5)
fix the issue

Reviewed 3 files

found the issue. The Jira API v3's `/project/search` endpoint returns projects in a `values` array, not a `projects` array. Let me fix the client:

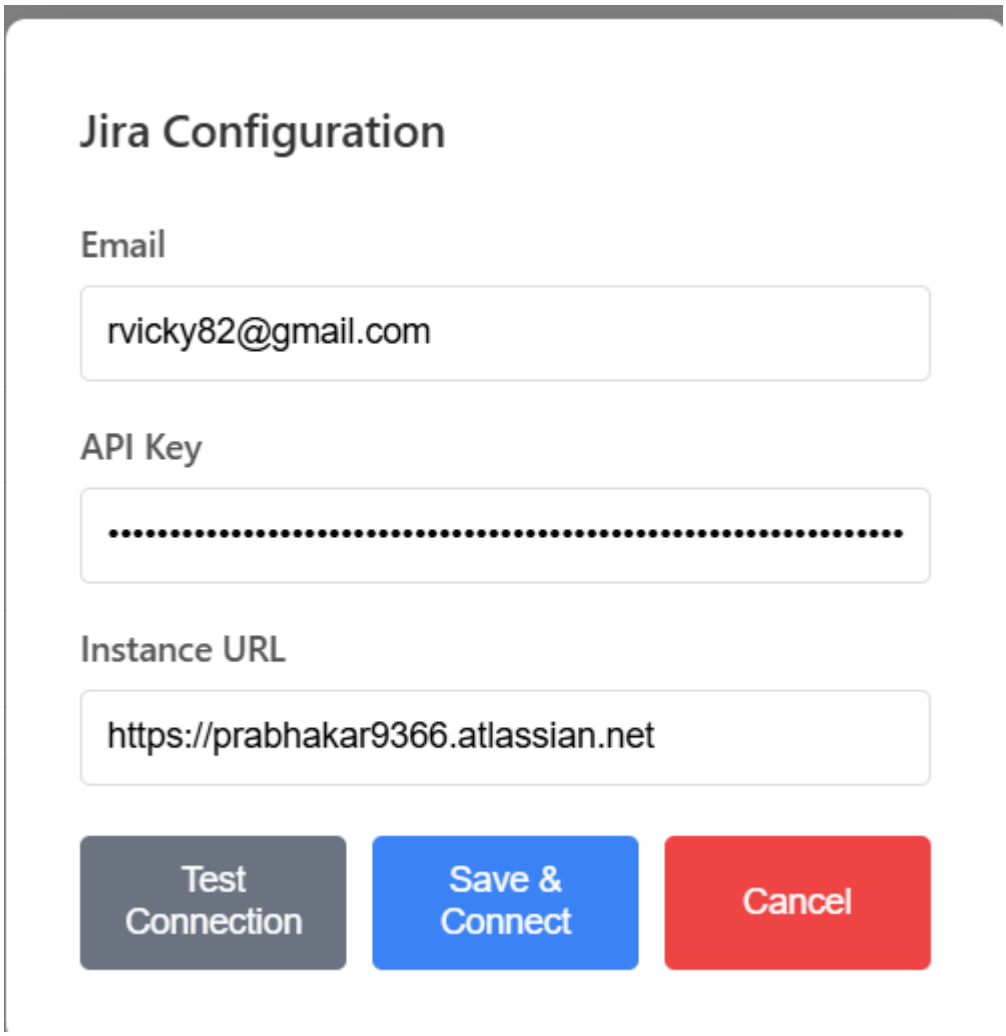
Updated jiraClient.ts and reviewed lines 80 to 120

```
found 0 vulnerabilities
PS D:\Test Leaf\user-story-to-tests (2)\user-story-to-tests\backend> npm run dev

> backend@1.0.0 dev
> tsx watch src/server.ts

Loading .env from: D:\Test Leaf\user-story-to-tests (2)\user-story-to-tests\.env
PS D:\Test Leaf\user-story-to-tests (2)\user-story-to-tests\backend> npm run dev

> backend@1.0.0 dev
> tsx watch src/server.ts

Loading .env from: D:\Test Leaf\user-story-to-tests (2)\user-story-to-tests\.env
Environment variables loaded:
PORT: 8091
CORS ORIGIN: http://localhost:5173
```

```
    at Module._load (node:internal/modules/cjs/loader:1335:12)
Emitted 'error' event on Server instance at:
    at process.processTicksAndRejections (node:internal/process/task_queues:90:21) {
  code: 'EADDRINUSE',
  errno: -4091,
  syscall: 'listen',
  address: '::',
  port: 8091
}
```

```
PS D:\Test Leaf\user-story-to-tests (2)\user-story-to-tests\frontend> npm install

up to date, audited 181 packages in 4s

32 packages are looking for funding
  run `npm fund` for details
```

```
PS D:\Test Leaf\user-story-to-tests (2)\user-story-to-tests\frontend> npm run dev

> frontend@0.0.0 dev
> vite


  VITE v5.4.21  ready in 1424 ms

  ➜  Local:   http://localhost:5173/
  ➜  Network: http://192.168.0.133:5173/
```



User Story to Tests

Generate comprehensive test cases from your user stories

Jira Integration

✓ Jira Settings

Jira Project

Select a project...

Story Title *

Task 3

Description

Additional description (optional)...

Acceptance Criteria *

Generate comprehensive test cases from your user stories

Jira Integration

✓ Jira Settings

Jira Project

My Project Management Team (SCRUM)

User Story

SCRUM-3: Task 3

Status: To Do
Reporter: Prabhakar P

Story Title *

Task 3

Description

Acceptance Criteria *

```
Story: SCRUM-3
Status: To Do
```

Additional Info

Any additional information (optional)...

Generate

Generated Test Cases

20 test case(s) generated • Model: meta-llama/llama-4-scout-17b-16e-instruct • Tokens: 2651

Test Case ID	Title	Category	Priority	Expected Result
TC-001	Verify SCRUM-3 story status update to In Progress	Positive	High	SCRUM-3 story status updated to In Progress
TC-002	Verify SCRUM-3 story status update to Done	Positive	High	SCRUM-3 story status updated to Done
TC-003	Verify SCRUM-3 story status update to To Do	Positive	High	SCRUM-3 story status updated to To Do
TC-004	Verify SCRUM-3 story status update with invalid status	Negative	Medium	Error message displayed for invalid status
TC-005	Verify SCRUM-3 story status update without authentication	Negative	Medium	Access denied error message displayed
TC-006	Verify SCRUM-3 story status update with unauthorized user	Negative	Medium	Access denied error message displayed
TC-007	Verify SCRUM-3 story status update with duplicate status	Negative	Low	Error message displayed for duplicate status
TC-007	Verify SCRUM-3 story status update with duplicate status	Negative	Low	Error message displayed for duplicate status
TC-008	Verify SCRUM-3 story status update with empty status	Negative	Low	Error message displayed for empty status
TC-009	Verify SCRUM-3 story status update with null status	Negative	Low	Error message displayed for null status
TC-010	Verify SCRUM-3 story status update with special characters	Negative	Low	Error message displayed for special characters
TC-011	Verify SCRUM-3 story status update with long status	Negative	Low	Error message displayed for long status
TC-012	Verify SCRUM-3 story status display	Positive	High	SCRUM-3 story status displayed
TC-013	Verify SCRUM-3 story status update notification	Positive	Medium	Notification sent for status update
TC-014	Verify SCRUM-3 story status history	Positive	Medium	SCRUM-3 story status history displayed
TC-015	Verify SCRUM-3 story status update with multiple users	Positive	Medium	SCRUM-3 story status updated with multiple users
TC-016	Verify SCRUM-3 story status update with multiple updates	Positive	Medium	SCRUM-3 story status updated multiple times
TC-017	Verify SCRUM-3 story status update with invalid date	Negative	Low	Error message displayed for invalid date
TC-015	Verify SCRUM-3 story status update with multiple users	Positive	Medium	SCRUM-3 story status updated with multiple users
TC-016	Verify SCRUM-3 story status update with multiple updates	Positive	Medium	SCRUM-3 story status updated multiple times
TC-017	Verify SCRUM-3 story status update with invalid date	Negative	Low	Error message displayed for invalid date
TC-018	Verify SCRUM-3 story status update with future date	Negative	Low	Error message displayed for future date
TC-019	Verify SCRUM-3 story status update with past date	Negative	Low	Error message displayed for past date
TC-020	Verify SCRUM-3 story status update with same status	Edge	Low	No change in SCRUM-3 story status

Export Test Cases

JSON CSV Excel

Files will be downloaded with today's date in the filename